Assemble and Disassemble a PC

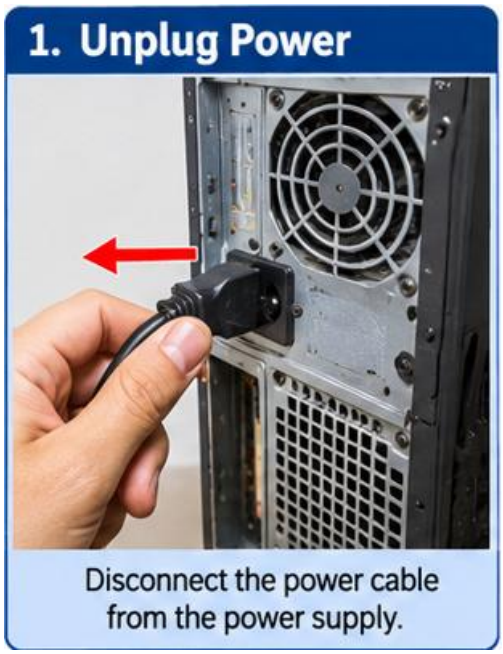
1.List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Component	Description
1. Motherboard	The main circuit board that connects all computer components and allows them to communicate.
2. CPU (Processor)	Known as the brain of the computer. It performs calculations and executes instructions.
3. RAM (Memory)	Temporarily stores data and programs that are currently being used by the computer.
4. HDD (Hard Disk Drive)	Stores data, files, software and the operating system permanently.
5. SSD (Solid State Drive)	Provides faster data storage and helps the computer start and load applications quickly.
6. GPU (Graphics Card)	Processes images, videos and graphics for display on the monitor.
7. SMPS (Power Supply)	Supplies electrical power to the motherboard, CPU, storage drives and other components.
8. CPU Cooling Fan	Keeps the processor cool and prevents overheating.
9. Heat Sink	Absorbs and transfers heat away from the CPU or other components.

10. Cabinet Fans	Maintain airflow inside the cabinet and remove hot air.
11. Optical Drive	Reads and sometimes writes CDs and DVDs.
12. CMOS Battery	Maintains the system clock and BIOS/UEFI settings when the computer is switched off.
13. Network Card (NIC)	Allows the computer to connect to a wired or wireless network.
14. Sound Card	Processes audio signals and provides sound output through speakers or headphones.
15. Expansion Slots	Allow additional cards, such as graphics, sound and network cards, to be connected to the motherboard.
16. SATA Cables	Connect storage drives such as HDDs and SATA SSDs to the motherboard.

**2.Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

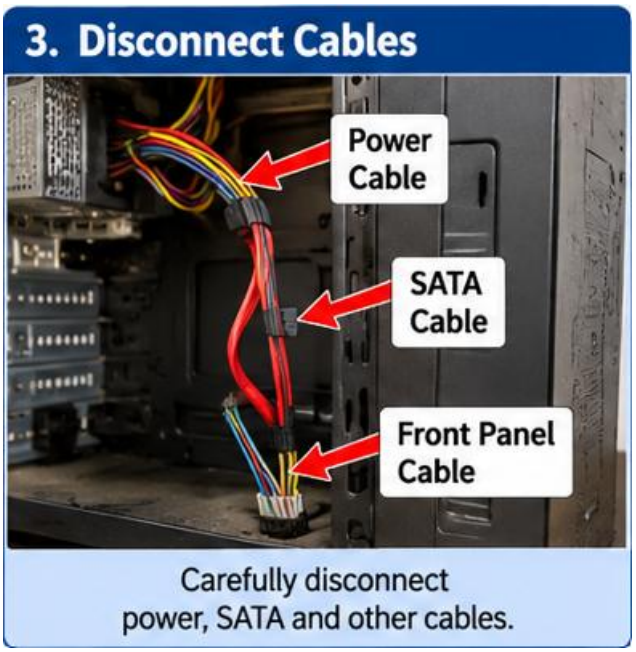
Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard. **



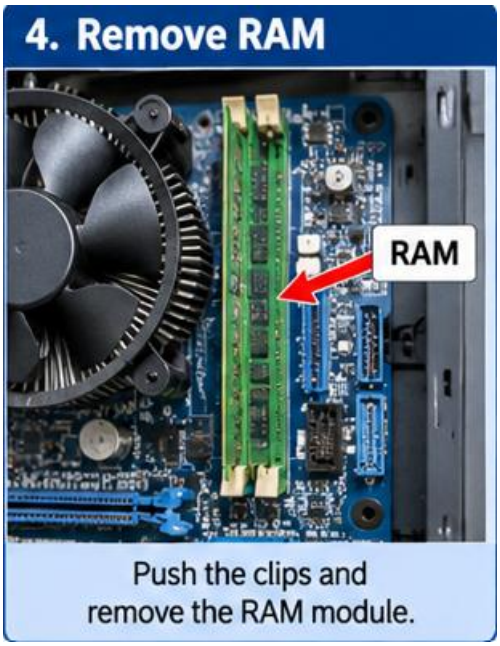
Caption: I shut down the PC and disconnected its power cable from the wall and the computer.



Caption: I removed the case screws and carefully slid the side panel away.



Caption: I took a reference photo, identified the connectors, and disconnected cables by holding the connector ends.



Caption: I released the RAM retaining clips and removed the module by holding its edges.



Caption: I disconnected the drive's power and data cables and removed the HDD/SSD from its bay or bracket.



Caption: After checking that all cables and attached components were disconnected, I removed the motherboard screws and lifted the board carefully.



Caption: I kept screws and components organized on an anti-static surface.

3.Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Step	Reassembly Process	Description
1	Install Motherboard	Place the motherboard inside the cabinet and secure it with screws. It provides connections for all components.
2	Install CPU	Carefully place the CPU into its socket and lock it properly. It performs all processing operations.
3	Install CPU Cooler	Attach the heatsink and cooling fan to prevent the processor from overheating.

4	Install RAM	Insert the RAM modules into the motherboard's memory slots until they lock into place.
5	Install HDD/SSD	Place the storage drive in its bay and secure it with screws. It stores the operating system and files.
6	Install GPU	Insert the graphics card into the PCIe slot and secure it to the cabinet.
7	Connect Power Cables	Connect the 24-pin motherboard cable, CPU power cable and GPU power cable if required.
8	Connect Data Cables	Connect SATA cables between the motherboard and storage drives.
9	Connect Cabinet Fans	Connect the cooling fans to the appropriate motherboard headers or power connectors.
10	Close Cabinet	Carefully attach the side panel and tighten the screws.
11	Power On and Test	Connect the power cable and switch on the PC to check whether it boots successfully.

4.Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

Safety Precautions Checklist:

No.	Safety Precaution	Why It Is Important

1	Switch off the computer and unplug the power cable.	Prevents electric shock and damage to components.
2	Remove all connected cables and devices.	Makes the work area safe and prevents cable damage.
3	Use an anti-static wrist strap.	Prevents static electricity from damaging components.
4	Work on a clean, dry and flat surface.	Prevents dust, moisture and accidental damage.
5	Handle components carefully by their edges.	Protects delicate circuits and connectors.
6	Keep screws and small parts safely in a container.	Prevents losing parts during assembly.
7	Do not force components into slots.	Prevents damage to RAM, CPU and other connectors.
8	Keep food and liquids away from the computer.	Prevents spills and short circuits.
9	Take photos before disconnecting cables.	Helps reconnect everything correctly.
10	Follow the manufacturer's instructions.	Ensures safe and correct installation.

Tools Required:

No.	Tool	Why It Is Important
1	Phillips-head screwdriver	Used to remove and tighten screws.
2	Anti-static wrist strap	Protects components from static electricity.

3	Anti-static mat	Provides a safe surface for handling components.
4	Small parts container	Keeps screws and small parts organized.
5	Cable ties	Helps organize cables inside the cabinet.
6	Compressed air	Removes dust from fans and other components.
7	Soft cleaning brush	Helps clean dust carefully from components.
8	Flashlight	Helps see small connectors and screws clearly.